

SANDSTONE 6 W, MN, USAWMO: **726460**Lat: **46.1135N**Lon: **92.9936W**Elev: **344**StdP: **97.26**Time Zone: **-6.00 (NAC)**Period: **07-19**WBAN: **54932****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.7	-24.9	-31.3	0.2	-27.3	-28.9	0.3	-24.4	4.7	-10.6	4.2	-10.7	0.6	N/A	0.493

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	29.5	21.7	28.0	20.5	26.7	19.5	23.4	27.9	22.1	26.4	21.0	25.1	2.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.9	17.3	26.6	20.6	15.9	24.9	19.5	14.9	23.6	71.6	27.7	66.4	26.5	62.4	25.3	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.5	3.9	3.5	-32.1	32.2	3.0	1.3	-34.2	33.2	-36.0	34.0	-37.7	34.7	-39.8	35.7		
(5)			-31.2	25.8	2.3	1.3	-32.9	26.7	-34.2	27.4	-35.5	28.2	-37.1	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-11.8	-9.7	-1.9	4.9	12.0	17.0	20.0	18.7	14.8	7.5	-0.5	-8.5
	DBStd	12.30	7.77	6.80	7.25	5.06	4.57	3.39	2.86	2.88	4.39	4.66	6.00	6.70
	HDD10.0	2803	677	552	373	164	29	1	0	0	9	107	317	572
	HDD18.3	4919	935	786	628	402	204	63	16	32	122	337	566	831
	CDD10.0	1081	0	0	4	13	91	211	310	268	153	29	2	0
	CDD18.3	156	0	0	0	0	7	23	67	42	16	1	0	0
	CDH23.3	1483	0	0	1	3	129	264	583	351	140	13	0	0
	CDH26.7	298	0	0	0	0	35	51	121	66	24	1	0	0
Wind	WSAvg	1.5	1.5	1.6	1.6	1.9	1.8	1.5	1.3	1.1	1.2	1.4	1.5	1.5
Precipitation	PrecAvg	802	23	23	37	72	81	117	110	103	83	81	46	28
	PrecMax	1007	63	64	67	205	166	191	179	235	154	170	132	59
	PrecMin	529	0	2	17	16	28	62	24	25	20	20	2	7
	PrecStd	126	17	17	16	42	32	34	42	50	43	41	38	16
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.6	9.3	20.6	23.0	30.4	30.6	31.5	30.5	29.4	25.4	17.8	6.9
		MCWB	2.4	6.1	14.7	13.3	17.9	21.6	24.6	23.2	21.1	15.8	11.5	4.9
	2%	DB	2.7	4.3	15.0	20.0	27.0	28.0	29.4	28.6	26.9	21.5	14.0	3.9
		MCWB	0.8	1.6	10.0	10.7	16.2	19.4	22.7	21.9	20.3	13.5	9.5	2.3
	5%	DB	0.9	2.0	10.9	17.1	23.9	26.3	28.0	26.8	24.9	17.9	10.7	1.8
		MCWB	-0.2	-0.1	6.8	9.1	14.9	18.1	21.1	20.2	18.4	11.7	6.9	0.8
	10%	DB	-0.9	-0.1	7.5	14.5	21.4	24.7	26.6	25.3	22.7	15.1	8.1	0.5
		MCWB	-2.0	-1.6	4.5	7.6	13.4	17.4	20.1	19.2	17.5	10.8	5.3	-0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.8	6.2	16.2	14.1	20.2	23.8	26.4	24.2	23.5	17.0	12.7	6.5
		MCDB	5.2	10.6	21.9	22.3	27.6	28.8	30.8	29.0	27.4	24.3	15.9	7.2
	2%	WB	1.3	2.8	11.0	11.5	18.1	21.6	24.0	22.6	21.5	14.9	9.4	2.8
		MCDB	2.6	4.9	14.1	17.9	23.2	26.5	28.2	27.1	25.2	19.6	13.4	4.0
	5%	WB	0.0	0.3	7.2	9.5	16.7	19.8	22.6	21.4	20.0	13.1	7.3	1.3
		MCDB	1.4	2.1	10.9	16.0	21.7	24.3	27.0	25.4	23.4	17.7	10.6	2.2
	10%	WB	-1.8	-1.3	4.5	7.8	15.2	18.5	21.3	20.3	18.5	11.4	5.4	0.2
		MCDB	-0.6	0.2	7.7	13.5	20.3	22.8	25.5	23.9	22.3	14.9	8.0	1.1
Mean Daily Temperature Range	5% DB	MDBR	9.4	10.3	10.6	11.3	13.1	11.9	11.4	11.5	11.6	9.9	8.3	7.7
		MCDBR	9.6	10.7	13.5	17.2	17.5	14.3	12.4	13.0	14.0	14.9	12.7	7.6
		MCWBR	7.9	7.6	8.6	9.4	8.4	7.5	6.5	6.9	7.4	8.5	8.5	6.4
	5% WB	MCDBR	9.3	10.2	12.8	15.1	14.0	12.5	11.5	11.6	11.7	13.4	12.3	6.8
MCWBR		7.8	7.5	8.8	9.0	8.0	7.4	6.8	6.9	7.1	8.1	8.4	5.8	
Clear-Sky Solar Irradiance	taub	0.250	0.261	0.307	0.362	0.391	0.379	0.378	0.389	0.362	0.311	0.284	0.258	
	taud	2.399	2.376	2.371	2.330	2.293	2.356	2.384	2.369	2.418	2.555	2.517	2.427	
	Ebn at Noon	873	930	923	895	876	886	882	858	853	862	825	819	
	Edh at Noon	69	87	103	118	127	120	115	113	98	74	63	61	
All-Sky Solar Radiation	RadAvg	1.63	2.74	3.84	4.65	5.33	5.85	6.21	5.18	3.95	2.44	1.53	1.14	
	RadStd	0.18	0.24	0.35	0.44	0.42	0.41	0.23	0.39	0.28	0.32	0.15	0.12	